


[Shop](#) [Learn](#) [Forums](#) [Bluesky](#) [Instagram](#) [Discord](#) [Github](#) [GBP](#) [Register / Login](#)

 Top notch support
57,036 reviews!

 Next day delivery
order by 2pm

 No fuss returns
contact support

 Secure shopping
3000+ products

PIMORONI

Search...


 £0
0 items

[Raspberry Pi](#) [Pico](#) [Pimoroni](#) [Adafruit](#) [Arduino & Co](#) [Breakouts](#) [Smart Home](#) [Kits](#) [Tools](#) [New](#) [D](#)

 Raspberry Pi 5 **NEW!** [Displays](#) [Cameras](#) [LED Strips](#) [Sensors](#) [Audio](#) [Feather](#) [Teensy](#) [Robotics](#) [Retro/Arcade](#)

Type here to search...


PIMORONI

LTD Purveyors of Maker Goods EST. 2012

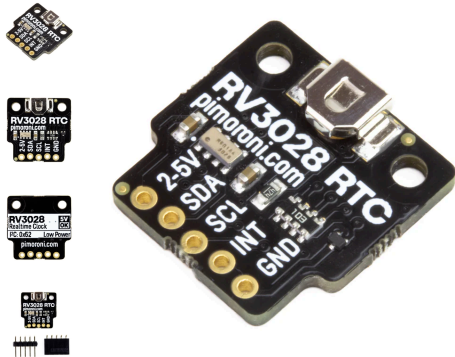
 £0
0 items

[Raspberry Pi](#) [Pico](#) [Pimoroni](#) [Adafruit](#) [Arduino & Co](#) [Breakouts](#) [Smart Home](#) [Kits](#) [Tools](#) [New](#) [D](#)

 Raspberry Pi 5 **NEW!** [Displays](#) [Cameras](#) [LED Strips](#) [Sensors](#) [Audio](#) [Feather](#) [Teensy](#) [Robotics](#) [Retro/Arcade](#)

RV3028 Real-Time Clock (RTC) Breakout

by Pimoroni



★★★★★ 14 reviews

login to add to wishlist

An ultra-low-power (~100 nA), highly accurate real-time clock breakout. The RV3028 RTC breakout is perfect for adding timekeeping to your project and, thanks to the tiny on-board battery, it'll keep time when your device is powered off. Like all the best timepieces, it's Swiss-made!


In stock – and ready to ship!

An ultra-low-power (~100 nA), highly accurate real-time clock breakout. The RV3028 RTC breakout is perfect for adding timekeeping to your project and, thanks to the tiny on-board battery, it'll keep time

RV3028 Real-Time Clock (RTC) Breakout

PIM449

£12.90

Add to cart

You can adjust quantities once added to your cart.

 Bulk or custom order? Go to our [wholesale store!](#)

Shop with confidence – we've been serving the hobbyist electronics, Maker, and retro gaming communities since 2012.

- Satisfaction or refund guarantee
- Worldwide shipping via mail or courier

when your device is powered off. Like all the best timepieces, it's Swiss-made!

The RV3028 RTC is factory-calibrated, and is quoted to drift by just ± 1 second per million seconds at 25 degrees. This will vary slightly depending on a number of factors. In our testing, we've seen performance better than ± 2 seconds per million seconds. For comparison, that's up to five times more accurate than your average digital watch.

The tiny silver oxide battery (8mAh, included) keeps the RTC running when your device (Raspberry Pi or Arduino) is powered off and can sync the time when powered back on. It should last for several years (up to nine in theory, although the battery shelf-life is five years)!

The RV3028 breakout has an I2C interface and is 3.3V or 5V compatible. Like our other [Pimoroni breakouts](#), we've designed it so that you can solder a piece of right-angle header onto it and then pop it straight onto the bottom left 5 pins on your Raspberry Pi's GPIO header (pins 1, 3, 5, 6, 9).

It's also compatible with our fancy [Breakout Garden](#), where using breakouts is as easy as just popping it into one of the six slots and starting to grow your project, create, and code.

- 57,000+ customer reviews
- Secure website and payments

"Say kids, what time is it ? Ah - that was easy !"



Anonymous

Features

- RV3028 Real-Time Clock (RTC)

- Ultra-low-power (~100nA typical current draw)
- ± 1 second drift per million seconds at 25 degrees
- **Silver oxide battery included (1.55V, 8mAh, 337 type)**
- 3.3V or 5V compatible
- I2C interface (address 0x52)
- Reverse polarity protection
- Raspberry Pi-compatible pinout (pins 1, 3, 5, 7, 9)
- Compatible with Arduino
- Compatible with all 40-pin Raspberry Pi models ([Python library](#))
- Compatible with Raspberry Pi Pico ([C++/MicroPython libraries](#))
- [Datasheet](#) and [applications manual](#)
- [Schematic](#)

Kit includes

- RV3028 RTC Breakout
- 1x5 pin header
- 1x5 right angle socket header

Software

We've put together a [Python library](#) that you can use with your RV3028 RTC Breakout, and an easy one-line installer to install everything.

Our [Python library](#) allows you to set and get the time (year, month, day, hour, minute, second), set periodic countdown timers with interrupts, and to set one-off or

recurring alarms with interrupts.

You can also use this breakout with Raspberry Pi Pico and other RP2040 boards, using [C++ or Pirate brand MicroPython](#).

Notes

- You'll need to run the [set-time.py](#) example the first time that you use your RV3028 RTC breakout to set the time correctly and to enable the clock to run on battery backup!
- The battery in this breakout is extremely tiny, so **take care not to lose it**. Also **take care not to contact any metal objects with the breakout** to avoid shorting the battery (don't toss it in a box with other breakouts or conductive materials)
- Dimensions: 19x19x4mm

14 customer reviews



2 years ago

That takes timekeeping to the next level. With ultra-low power consumption, high accuracy, and a user-friendly breakout board, the TimeSync Master ensures precise timekeeping for your projects. Its compact design and simple interface make it ideal for both beginners and experienced makers. The breakout board includes convenient connectors and clear labeling, streamlining the integration of the RV3028 RTC into your electronic setups. Stay synced and on time with the TimeSync Master Breakout



by Aitor about RV3028 Real-Time Clock (RTC) Breakout via REVIEWS.io



2 years ago

Awesome piece of kit, very small. its going into my tufty badge along with a lipo charger :) Thank you for excellent service



by Cassiopeia about [RV3028 Real-Time Clock \(RTC\) Breakout](#) via [REVIEWS.io](#)



3 years ago



by Daniel about [RV3028 Real-Time Clock \(RTC\) Breakout](#) via [REVIEWS.io](#)



3 years ago

Works as intended. The example code provided on github allows for some easy alterations in ways times and dates are displayed



by Timothy about [RV3028 Real-Time Clock \(RTC\) Breakout](#) via [REVIEWS.io](#)



3 years ago

Quick and easy, accurate and (relatively) inexpensive. Not as cheap as some of the Chinese specials you can get elsewhere, but a lot simpler to use. I love the breakout garden range -



by Anonymous about [RV3028 Real-Time Clock \(RTC\) Breakout](#) via [REVIEWS.io](#)



4 years ago

Say kids, what time is it ? Ah - that was easy !



by Anonymous about [RV3028 Real-Time Clock \(RTC\) Breakout](#) via [REVIEWS.io](#)



4 years ago

Really simple to use and the board allows you to use it in the breakout boards or simply soldering connectors onto it. Great for use with the Pico to



by Erick about [RV3028 Real-Time Clock \(RTC\) Breakout](#) via [REVIEWS.io](#)



4 years ago



by Zoe about [RV3028 Real-Time Clock \(RTC\) Breakout](#) via [REVIEWS.io](#)

**4 years ago**

The module looks well made. I haven't actually tested it yet but I can't see why it wouldn't work. It came with a battery installed although it seemed to be an old one. Perhaps used for testing and left in there? One odd feature is the 470K series resistor in line with the battery. Technical support is via a forum that doesn't appear to get much attention from Pimoroni staff. It would be nice if schematics were available

**4 years ago**

Nice little RTC breakout board. I began playing around. I use it with MicroPython and can for now set time/date and reread after a moment. I use it with a Pico connected via USB to a



Real Time Clock (RTC)
Breakout via [REVIEWS.io](#)

**5 years ago**

by Richard about [RV3028](#)
[Real-Time Clock \(RTC\)](#)
[Breakout](#) via [REVIEWS.io](#)

**5 years ago**

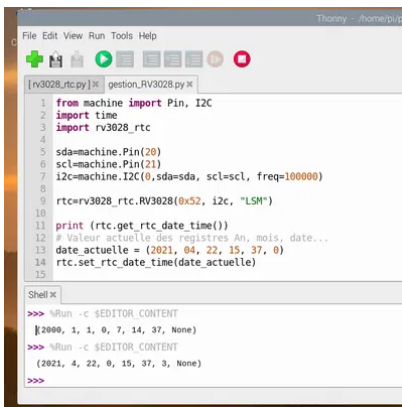
Works well, wide voltage range
is a life saver



by Timothy about [RV3028](#)
[Real-Time Clock \(RTC\)](#)
[Breakout](#) via [REVIEWS.io](#)

[Show more reviews...](#)

3 user photos



SUBSCRIBE FOR UPDATES

We'll send you information about new product

THE GOOD SHIP PIMORONI

We've been sailing the Maker seas

INFORMATION

Not sure about something? Check

FREQUENTLY ASKED QUESTIONS

Often it's the simplest of things.

launches and
special offers!

Enter your e-m

Subscribe

offering our wares
since 2012.

- [read about our history](#)
- [what our customers say](#)
- [check out our products](#)

Follow us on
social media:
[Bluesky](#), [YouTube](#),
[Facebook](#),
[Instagram](#).

out the links
below.

- [shipping information](#)
- [ask a question in our forums](#)
- [contact our support team](#)
- [find a distributor](#)

- [how can I track my order?](#)
- [cost and time to ship to ...?](#)
- [are you VAT registered in the EU?](#)
- [can I collect my order?](#)

View other
[frequently asked questions](#).

PIMORONI LTD.

Unit 1, Parkway Business Park, Parkway Drive, Sheffield, S9 4WN, United Kingdom

Registered in England and Wales Company №: 07510759 VAT Reg №: GB 140 8707 21

[Terms & Conditions](#) | [Privacy Policy](#) | [Cellular Service T&Cs](#)